Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Souza Brothers Dairy

Physical address of dairy:

3887 McCullagh RD	Stevinson	Merced	95374
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X055-X120-X045-XXXX

B. OPERATORS

Souza, Joe

Operator name: Souza, Joe

Telephone no.: (209) 632-3773 (Landline) (Cellular)

3325 McCullagh RD	Stevinson	CA	95374
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

Souza, John

Operator name: Souza, John

Telephone no.: (209) 632-3373 (Landline) (209) 634-7522 (Cellular)

3325 McCullagh RD	Stevinson	CA	95374
Mailing Address Number and Street	City	State	Zip Code

C. OWNERS

Souza, Joe

Legal owner name: Souza, Joe

Telephone no.: (209) 632-3773
Landline | Cellular

3325 McCullagh RD	Stevinson	CA	95374
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

Souza, John

Legal owner name: Souza, John

Telephone no.: (209) 632-3373 (209) 634-7522
Landline | Cellular

3325 McCullagh RD	Stevinson	CA	95374
Mailing Address Number and Street	City	State	Zip Code

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	462	0	0	0
Number under roof	560	95	0	120	205	0
Maximum number	560	95	462	120	205	0
Average number	260	95	462	120	205	0
Avg live weight (lbs)	1,400	1,500	1,100	775		

Predominant milk cow breed: Holstein

Average milk production: 73 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 15,026.44 *tons per reporting period*

Total nitrogen from manure: 171,440.50 *lbs per reporting period*

After ammonia losses (30% loss applied): 120,008.35 *lbs per reporting period*

Total phosphorus from manure: 27,358.55 *lbs per reporting period*

Total potassium from manure: 50,304.86 *lbs per reporting period*

Total salt from manure: 144,266.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 10,380,000 *gallons*

Total nitrogen generated: 59,553.86 *lbs*

Total phosphorus generated: 10,529.80 *lbs*

Total potassium generated: 63,410.78 *lbs*

Total salt generated: 392,809.16 *lbs*

	10,380,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	10,380,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - Dairy Well DW	Ground water
Domestic Well - Dry Stock DW	Ground water
Domestic Well - South Pump DW	Ground water
Irrigation Water - Canal	Surface water
Irrigation Water - River Water	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	Corral solids	1,100.00 *ton*	Dry-weight	11.7		20,100.00	7,900.00	22,100.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	39,046.26	15,346.54	42,931.46	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	39,046.26	15,346.54	42,931.46	0.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	38	38	7	manure	X055-X009-X005-XXXX X055-X012-X045-XXXX
10	37	37	2	both	X055-X010-X011-XXXX X055-X011-X038-XXXX
11	63	63	2	manure	X055-X010-X020-XXXX
12	35	35	7	manure	X055-X010-X020-XXXX
13	18	18	1	manure	X055-X010-X020-XXXX
14	60	60	7	manure	X055-X010-X020-XXXX
2	42	42	1	both	X055-X012-X045-XXXX
3	31	31	0	none	X055-X012-X046-XXXX X055-X120-X046-XXXX
4	32	32	0	none	X055-X012-X046-XXXX
5	33	33	0	none	X055-X012-X046-XXXX
6	44	44	0	none	X055-X012-X046-XXXX
7	57	57	2	both	X055-X012-X001-XXXX
8	61	61	2	process wastewater	X055-X009-X006-XXXX X055-X010-X011-XXXX X055-X012-X001-XXXX
9	43	43	2	both	X055-X011-X038-XXXX
Totals for areas that were used for application	454	454	33		
Totals for areas that were not used for application	140	140	0		
Land application area totals	594	594	33		

B. CROPS AND HARVESTS

1

Field name: 1

1

11/06/2018: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 38 Plant date: 11/06/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	89.00 *ton*	Dry-weight		10.5	21,100.00	3,400.00	18,500.00		7.60
05/26/2023	42.00 *ton*	Dry-weight		37.4	36,000.00	3,100.00	17,500.00		9.90
06/24/2023	43.00 *ton*	Dry-weight		10.5	21,100.00	3,400.00	18,500.00		7.60
07/30/2023	42.00 *ton*	Dry-weight		10.5	21,100.00	3,400.00	18,500.00		7.60
08/23/2023	44.00 *ton*	Dry-weight		10.5	21,100.00	3,400.00	18,500.00		7.60
09/20/2023	43.00 *ton*	Dry-weight		10.5	21,100.00	3,400.00	18,500.00		7.60
10/19/2023	40.00 *ton*	Dry-weight		10.5	21,100.00	3,400.00	18,500.00		7.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	9.03	348.99	52.50	286.52	1,214.58

10

Field name: 10

11/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 11/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	487.00 *ton*	Dry-weight		55.4	12,500.00	2,300.00	19,200.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.16	146.76	27.00	225.42	904.03

10

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,210.00 *ton*	Dry-weight		73.8	13,100.00	2,900.00	13,300.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.70	224.48	49.70	227.91	891.08

11

Field name: 11

11/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 63 Plant date: 11/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	958.00 *ton*	Dry-weight		59.3	11,300.00	2,000.00	19,100.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.21	139.87	24.76	236.42	953.10

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 63 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	2,058.00 *ton*	Dry-weight		73.8	11,000.00	2,500.00	14,900.00		5.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.67	188.29	42.79	255.05	907.22

12

Field name: 12

11/06/2018: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 35 Plant date: 11/06/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	116.00 *ton*	Dry-weight		10.8	20,900.00	3,400.00	19,800.00		8.30
05/26/2023	33.00 *ton*	Dry-weight		10.8	20,900.00	3,400.00	19,800.00		8.30
06/24/2023	34.00 *ton*	Dry-weight		10.8	20,900.00	3,400.00	19,800.00		8.30
07/30/2023	35.00 *ton*	Dry-weight		10.8	20,900.00	3,400.00	19,800.00		8.30
08/23/2023	35.00 *ton*	Dry-weight		10.8	20,900.00	3,400.00	19,800.00		8.30
09/20/2023	30.00 *ton*	Dry-weight		10.8	20,900.00	3,400.00	19,800.00		8.30
10/19/2023	32.00 *ton*	Dry-weight		10.8	20,900.00	3,400.00	19,800.00		8.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	9.00	335.57	54.59	317.91	1,332.65

13

Field name: 13

11/23/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 11/23/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	280.00 *ton*	Dry-weight		59.5	9,400.00	1,900.00	18,600.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.56	118.44	23.94	234.36	781.20

14

Field name: 14

14

11/11/2016: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 60 Plant date: 11/11/2016

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	189.00 *ton*	Dry-weight		10.3	22,700.00	3,700.00	20,300.00		8.70
05/26/2023	66.00 *ton*	Dry-weight		10.3	22,700.00	3,700.00	20,300.00		8.70
06/24/2023	64.00 *ton*	Dry-weight		10.3	22,700.00	3,700.00	20,300.00		8.70
07/30/2023	60.00 *ton*	Dry-weight		10.3	22,700.00	3,700.00	20,300.00		8.70
08/23/2023	62.00 *ton*	Dry-weight		10.3	22,700.00	3,700.00	20,300.00		8.70
09/20/2023	62.00 *ton*	Dry-weight		10.3	22,700.00	3,700.00	20,300.00		8.70
10/19/2023	61.00 *ton*	Dry-weight		10.3	22,700.00	3,700.00	20,300.00		8.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	9.40	382.80	62.40	342.33	1,467.13

2

Field name: 2

04/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 42 Plant date: 04/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,385.00 *ton*	Dry-weight		74.3	13,300.00	2,400.00	13,500.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.98	225.43	40.68	228.82	881.39

7

Field name: 7

7

11/21/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 57 Plant date: 11/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	789.00 *ton*	Dry-weight		51.3	11,400.00	2,100.00	19,500.00		6.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.84	153.70	28.31	262.90	822.41

05/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 57 Plant date: 05/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,832.00 *ton*	Dry-weight		74.5	13,000.00	2,700.00	12,900.00		4.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.14	213.09	44.26	211.45	803.19

8

Field name: 8

11/21/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 61 Plant date: 11/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	1,035.00 *ton*	Dry-weight		57.0	12,300.00	2,300.00	19,800.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.97	179.48	33.56	288.92	977.65

8

05/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 61 Plant date: 05/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	2,012.00 *ton*	Dry-weight		73.8	11,300.00	2,300.00	12,900.00		4.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.98	195.30	39.75	222.96	812.32

9

Field name: 9

11/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 43 Plant date: 11/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	536.00 *ton*	Dry-weight		53.8	12,800.00	2,200.00	19,300.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.47	147.43	25.34	222.29	794.73

05/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 43 Plant date: 05/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,398.00 *ton*	Dry-weight		74.4	12,100.00	2,400.00	13,800.00		4.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.51	201.42	39.95	229.71	815.65

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/06/2018: Alfalfa, hay

Field name: 1

Crop: Alfalfa, hay

Plant date: 11/06/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.17	0.00	0.00	39.21	3,798,445.00 *gal*
Application event totals		0.17	0.00	0.00	39.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.28	0.00	0.00	66.45	6,437,838.00 *gal*
Application event totals		0.28	0.00	0.00	66.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.21	0.00	0.00	49.29	4,775,368.00 *gal*
Application event totals		0.21	0.00	0.00	49.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.21	0.00	0.00	49.82	4,826,831.00 *gal*
Application event totals		0.21	0.00	0.00	49.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.20	0.00	0.00	47.10	4,562,892.00 *gal*
Application event totals		0.20	0.00	0.00	47.10	

1 - 11/06/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.26	0.00	0.00	60.27	5,839,250.00 *gal*
Application event totals		0.26	0.00	0.00	60.27	

11/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	298.92	117.49	328.66	0.00	320.00 *ton*
Application event totals		298.92	117.49	328.66	0.00	

10 - 11/23/2022: Oats, silage-soft dough

Field name: 10

Crop: Oats, silage-soft dough Plant date: 11/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	60.63	8.76	74.15	388.04	465,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	122.93	6,566,625.00 *gal*
Application event totals		60.63	8.76	74.15	510.97	

11/13/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	134.31	52.79	147.68	0.00	140.00 *ton*
Application event totals		134.31	52.79	147.68	0.00	

10 - 05/31/2023: Corn, silage

Field name: 10

Crop: Corn, silage Plant date: 05/31/2023

10 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	297.40	116.89	327.00	0.00	310.00 *ton*
Application event totals		297.40	116.89	327.00	0.00	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.23	0.00	0.00	54.20	5,112,928.00 *gal*
Application event totals		0.23	0.00	0.00	54.20	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	167.02	8,921,820.00 *gal*
Application event totals		0.00	0.00	0.00	167.02	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.17	0.00	0.00	40.40	3,810,827.00 *gal*
Application event totals		0.17	0.00	0.00	40.40	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.60	0.00	0.00	139.93	13,200,600.00 *gal*
Application event totals		0.60	0.00	0.00	139.93	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.62	0.00	0.00	145.64	13,739,400.00 *gal*
Application event totals		0.62	0.00	0.00	145.64	

11 - 11/23/2022: Oats, silage-soft dough

11 - 11/23/2022: Oats, silage-soft dough

Field name: 11

Crop: Oats, silage-soft dough

Plant date: 11/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	23.32	2,121,525.00 *gal*
Application event totals		0.00	0.00	0.00	23.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.16	0.00	0.00	36.58	5,876,397.00 *gal*
Application event totals		0.16	0.00	0.00	36.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/13/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	185.93	73.08	204.44	0.00	330.00 *ton*
Application event totals		185.93	73.08	204.44	0.00	

11 - 05/22/2023: Corn, silage

Field name: 11

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	253.55	99.65	278.78	0.00	450.00 *ton*
Application event totals		253.55	99.65	278.78	0.00	

11 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.44	0.00	0.00	104.08	16,718,763.00 *gal*
Application event totals		0.44	0.00	0.00	104.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.32	0.00	0.00	74.57	11,977,305.00 *gal*
Application event totals		0.32	0.00	0.00	74.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.24	0.00	0.00	57.33	9,208,549.00 *gal*
Application event totals		0.24	0.00	0.00	57.33	

12 - 11/06/2018: Alfalfa, hay

Field name: 12

Crop: Alfalfa, hay

Plant date: 11/06/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.18	0.00	0.00	41.30	3,685,701.00 *gal*
Application event totals		0.18	0.00	0.00	41.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.16	0.00	0.00	38.11	3,400,907.00 *gal*
Application event totals		0.16	0.00	0.00	38.11	

12 - 11/06/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.16	0.00	0.00	37.34	3,331,826.00 *gal*
Application event totals		0.16	0.00	0.00	37.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.21	0.00	0.00	49.84	4,447,866.00 *gal*
Application event totals		0.21	0.00	0.00	49.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.23	0.00	0.00	54.01	4,819,988.00 *gal*
Application event totals		0.23	0.00	0.00	54.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.24	0.00	0.00	55.47	4,949,677.00 *gal*
Application event totals		0.24	0.00	0.00	55.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	304.26	119.58	334.53	0.00	300.00 *ton*
Application event totals		304.26	119.58	334.53	0.00	

13 - 11/23/2022: Oats, silage-soft dough

Field name: 13

Crop: Oats, silage-soft dough

Plant date: 11/23/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

13 - 11/23/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	660.86	17,174,250.00 *gal*
Irrigation Water - River Water	Surface water	0.62	0.00	0.00	144.64	6,637,911.00 *gal*
Application event totals		0.62	0.00	0.00	805.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.26	0.00	0.00	61.26	2,811,442.00 *gal*
Application event totals		0.26	0.00	0.00	61.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	1.19	0.00	0.00	280.03	12,851,563.00 *gal*
Application event totals		1.19	0.00	0.00	280.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.71	0.00	0.00	166.00	7,618,396.00 *gal*
Application event totals		0.71	0.00	0.00	166.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/13/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	138.04	54.26	151.78	0.00	70.00 *ton*
Application event totals		138.04	54.26	151.78	0.00	

14 - 11/11/2016: Alfalfa, hay

Field name: 14

Crop: Alfalfa, hay

Plant date: 11/11/2016

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

14 - 11/11/2016: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.26	0.00	0.00	60.35	9,232,011.00 *gal*
Application event totals		0.26	0.00	0.00	60.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.23	0.00	0.00	54.08	8,273,683.00 *gal*
Application event totals		0.23	0.00	0.00	54.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.24	0.00	0.00	57.15	8,741,931.00 *gal*
Application event totals		0.24	0.00	0.00	57.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.24	0.00	0.00	56.85	8,697,289.00 *gal*
Application event totals		0.24	0.00	0.00	56.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	349.05	137.19	383.78	0.00	590.00 *ton*
Application event totals		349.05	137.19	383.78	0.00	

2 - 04/17/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 04/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

2 - 04/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	64.52	26.36	126.26	0.00	170.00 *ton*
Application event totals		64.52	26.36	126.26	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	235.19	56.49	181.32	1,069.35	1,170,000.00 *gal*
Irrigation Water - River Water	Surface water	0.44	0.00	0.00	102.61	10,987,696.00 *gal*
Application event totals		235.62	56.49	181.32	1,171.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	73.31	4,445,100.00 *gal*
Application event totals		0.00	0.00	0.00	73.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.17	0.00	0.00	40.85	4,373,898.00 *gal*
Application event totals		0.17	0.00	0.00	40.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.16	0.00	0.00	38.17	4,087,801.00 *gal*
Application event totals		0.16	0.00	0.00	38.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.16	0.00	0.00	37.23	3,986,461.00 *gal*
Application event totals		0.16	0.00	0.00	37.23	

2 - 04/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.18	0.00	0.00	43.40	4,647,287.00 *gal*
Application event totals		0.18	0.00	0.00	43.40	

7 - 11/21/2022: Oats, silage-soft dough

Field name: 7

Crop: Oats, silage-soft dough

Plant date: 11/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	57.77	13.87	44.54	262.65	390,000.00 *gal*
Irrigation Water - River Water	Surface water	0.03	0.00	0.00	7.22	1,049,892.00 *gal*
Application event totals		57.80	13.87	44.54	269.87	

02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.22	5.34	17.13	101.02	150,000.00 *gal*
Irrigation Water - River Water	Surface water	0.01	0.00	0.00	2.39	346,705.00 *gal*
Application event totals		22.23	5.34	17.13	103.40	

02/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	64.75	9.35	79.18	414.40	765,000.00 *gal*
Irrigation Water - River Water	Surface water	0.03	0.00	0.00	7.23	1,050,218.00 *gal*
Application event totals		64.78	9.35	79.18	421.62	

7 - 11/21/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	2.54	0.37	3.11	16.25	30,000.00 *gal*
Irrigation Water - River Water	Surface water	0.01	0.00	0.00	1.63	237,220.00 *gal*
Application event totals		2.55	0.37	3.11	17.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	31.88	3.54	43.60	287.77	420,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	38.06	3,131,775.00 *gal*
Application event totals		31.88	3.54	43.60	325.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.17	0.00	0.00	39.55	5,748,012.00 *gal*
Application event totals		0.17	0.00	0.00	39.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/13/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	27.97	11.43	54.73	0.00	100.00 *ton*
Application event totals		27.97	11.43	54.73	0.00	

7 - 05/19/2023: Corn, silage

Field name: 7

Crop: Corn, silage

Plant date: 05/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

7 - 05/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	286.46	112.59	314.97	0.00	460.00 *ton*
Application event totals		286.46	112.59	314.97	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.32	0.00	0.00	75.58	10,984,062.00 *gal*
Application event totals		0.32	0.00	0.00	75.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.25	0.00	0.00	59.06	8,582,899.00 *gal*
Application event totals		0.25	0.00	0.00	59.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.17	0.00	0.00	38.99	5,666,696.00 *gal*
Application event totals		0.17	0.00	0.00	38.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.17	0.00	0.00	40.25	5,849,025.00 *gal*
Application event totals		0.17	0.00	0.00	40.25	

8 - 11/21/2022: Oats, silage-soft dough

Field name: 8

Crop: Oats, silage-soft dough

Plant date: 11/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

8 - 11/21/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	28.73	3.19	39.28	259.30	405,000.00 *gal*
Irrigation Water - River Water	Surface water	0.02	0.00	0.00	3.89	605,757.00 *gal*
Application event totals		28.74	3.19	39.28	263.19	

01/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	36.18	4.02	49.47	326.52	510,000.00 *gal*
Irrigation Water - River Water	Surface water	0.03	0.00	0.00	6.80	1,057,386.00 *gal*
Application event totals		36.20	4.02	49.47	333.32	

02/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	24.47	2.72	33.46	220.88	345,000.00 *gal*
Irrigation Water - River Water	Surface water	0.02	0.00	0.00	3.80	590,768.00 *gal*
Application event totals		24.49	2.72	33.46	224.68	

02/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	47.88	5.32	65.47	432.16	675,000.00 *gal*
Irrigation Water - River Water	Surface water	0.05	0.00	0.00	12.60	1,959,342.00 *gal*
Application event totals		47.93	5.32	65.47	444.76	

03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	30.86	3.43	42.19	278.50	435,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.71	3,232,800.00 *gal*
Application event totals		30.86	3.43	42.19	315.21	

8 - 11/21/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	51.07	5.67	69.84	460.97	720,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	61.94	5,455,350.00 *gal*
Application event totals		51.07	5.67	69.84	522.92	

09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.07	0.00	0.00	16.57	2,576,830.00 *gal*
Application event totals		0.07	0.00	0.00	16.57	

09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	25.54	2.84	34.92	230.49	360,000.00 *gal*
Irrigation Water - River Water	Surface water	0.05	0.00	0.00	12.52	1,946,634.00 *gal*
Application event totals		25.59	2.84	34.92	243.00	

8 - 05/19/2023: Corn, silage

Field name: 8

Crop: Corn, silage

Plant date: 05/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.53	0.00	0.00	124.80	19,409,048.00 *gal*
Application event totals		0.53	0.00	0.00	124.80	

8 - 05/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	102.14	11.35	139.67	921.94	1,440,000.00 *gal*
Irrigation Water - River Water	Surface water	0.25	0.00	0.00	59.92	9,318,576.00 *gal*
Application event totals		102.40	11.35	139.67	981.86	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	151.55	36.40	116.84	689.08	1,095,000.00 *gal*
Irrigation Water - River Water	Surface water	0.26	0.00	0.00	61.64	9,586,882.00 *gal*
Application event totals		151.81	36.40	116.84	750.72	

09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.23	0.00	0.00	54.60	8,491,677.00 *gal*
Application event totals		0.23	0.00	0.00	54.60	

9 - 11/22/2022: Oats, silage-soft dough

Field name: 9

Crop: Oats, silage-soft dough

Plant date: 11/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.03	0.00	0.00	7.42	812,998.00 *gal*
Application event totals		0.03	0.00	0.00	7.42	

9 - 11/22/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.29	9.20	29.52	174.08	195,000.00 *gal*
Irrigation Water - River Water	Surface water	0.02	0.00	0.00	4.39	481,608.00 *gal*
Application event totals		38.31	9.20	29.52	178.47	

03/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	40.75	4.53	55.73	367.84	405,000.00 *gal*
Irrigation Water - River Water	Surface water	0.04	0.00	0.00	9.23	1,011,442.00 *gal*
Application event totals		40.79	4.53	55.73	377.07	

03/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	79.52	19.10	61.31	361.55	405,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.82	3,030,750.00 *gal*
Application event totals		79.52	19.10	61.31	410.37	

09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	80.65	5,007,026.00 *gal*
Application event totals		0.00	0.00	0.00	80.65	

11/13/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	40.78	16.66	79.80	0.00	110.00 *ton*
Application event totals		40.78	16.66	79.80	0.00	

9 - 05/29/2023: Corn, silage

Field name: 9

Crop: Corn, silage

Plant date: 05/29/2023

9 - 05/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	272.42	107.07	299.52	0.00	330.00 *ton*
Application event totals		272.42	107.07	299.52	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.33	0.00	0.00	77.29	8,473,420.00 *gal*
Application event totals		0.33	0.00	0.00	77.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.43	0.00	0.00	100.91	11,063,094.00 *gal*
Application event totals		0.43	0.00	0.00	100.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.17	0.00	0.00	39.87	4,371,451.00 *gal*
Application event totals		0.17	0.00	0.00	39.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.26	0.00	0.00	60.64	6,648,012.00 *gal*
Application event totals		0.26	0.00	0.00	60.64	

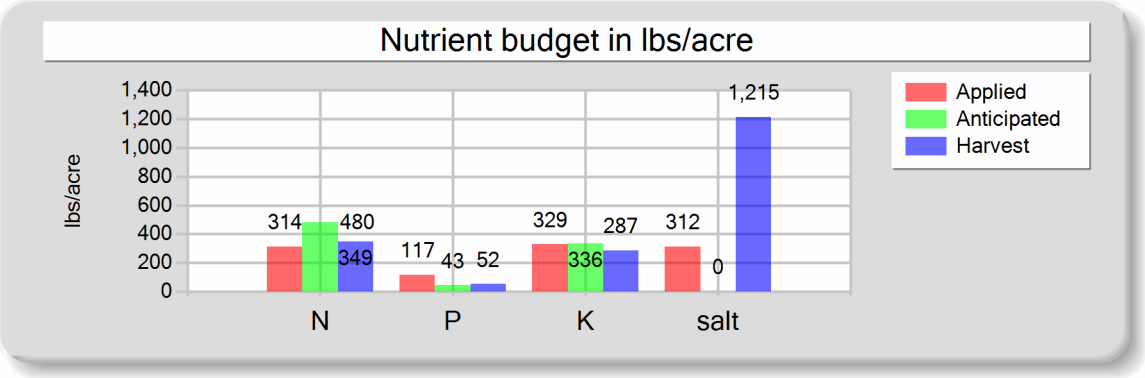
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - River Water	Surface water	0.23	0.00	0.00	55.02	6,031,502.00 *gal*
Application event totals		0.23	0.00	0.00	55.02	

B. NUTRIENT BUDGET

1 - 11/06/2018: Alfalfa, hay

Field name: 1 Crop: Alfalfa, hay Plant date: 11/06/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	298.92	117.49	328.66	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.33	0.00	0.00	312.13
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	314.25	117.49	328.66	312.13
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	348.99	52.50	286.52	1,214.58
Nutrient balance	-34.74	64.99	42.14	-902.45
Applied to removed ratio	0.90	2.24	1.15	0.26

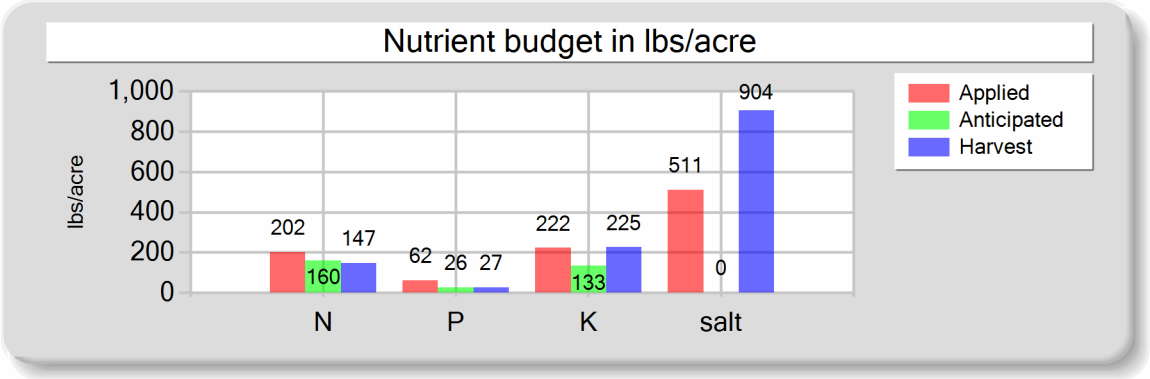
Fresh water applied	
30,240,624.00	*gallons*
1,113.66	*acre-inches*
29.31	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
7	*harvests*

10 - 11/23/2022: Oats, silage-soft dough

Field name: 10 Crop: Oats, silage-soft dough Plant date: 11/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	134.31	52.79	147.68	0.00
Process wastewater	60.63	8.76	74.15	388.04
Fresh water	0.00	0.00	0.00	122.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	201.94	61.55	221.82	510.97
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	146.76	27.00	225.42	904.03
Nutrient balance	55.18	34.54	-3.60	-393.06
Applied to removed ratio	1.38	2.28	0.98	0.57

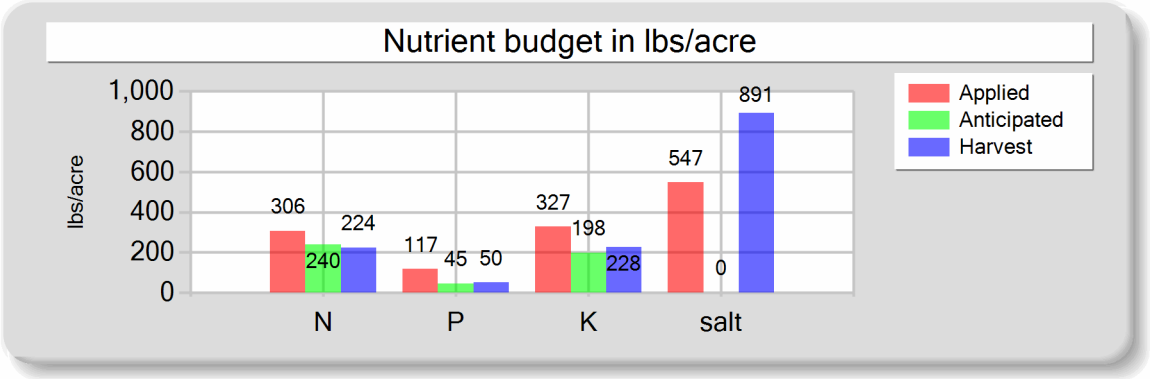
Fresh water applied	
6,566,625.00	*gallons*
241.83	*acre-inches*
6.54	*inches/acre*

Process wastewater applied	
465,000.00	*gallons*
17.12	*acre-inches*
0.46	*inches/acre*

Total harvests for the crop	
1	*harvests*

10 - 05/31/2023: Corn, silage

Field name: 10 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	297.40	116.89	327.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.62	0.00	0.00	547.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	306.02	116.89	327.00	547.19
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	224.48	49.70	227.91	891.08
Nutrient balance	81.54	67.20	99.08	-343.90
Applied to removed ratio	1.36	2.35	1.43	0.61

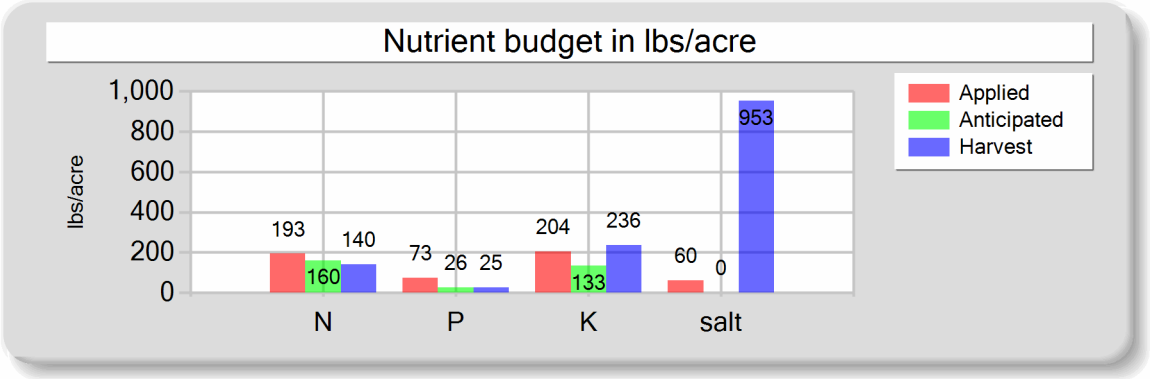
Fresh water applied	
44,785,575.00	*gallons*
1,649.30	*acre-inches*
44.58	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

11 - 11/23/2022: Oats, silage-soft dough

Field name: 11 Crop: Oats, silage-soft dough Plant date: 11/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	185.93	73.08	204.44	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.16	0.00	0.00	59.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	193.09	73.08	204.44	59.91
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	139.87	24.76	236.42	953.10
Nutrient balance	53.22	48.32	-31.98	-893.19
Applied to removed ratio	1.38	2.95	0.86	0.06

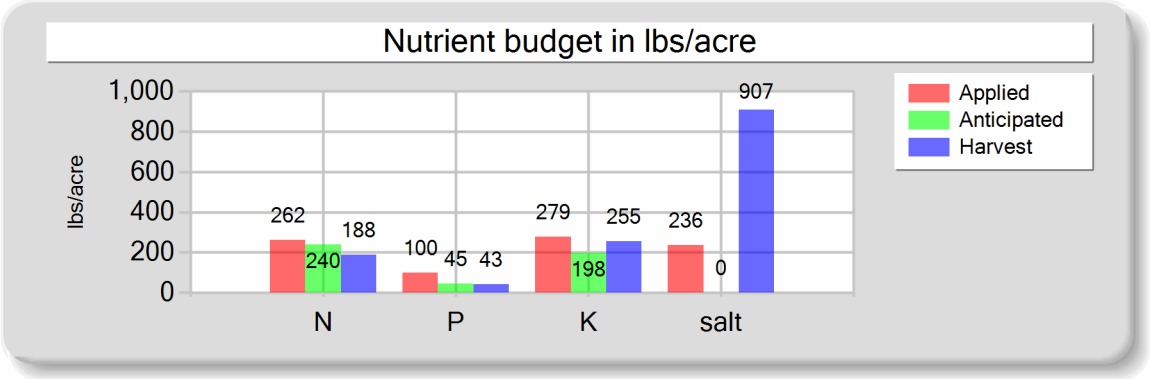
Fresh water applied	
7,997,922.00	*gallons*
294.54	*acre-inches*
4.68	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

11 - 05/22/2023: Corn, silage

Field name: 11 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	253.55	99.65	278.78	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.00	0.00	0.00	235.98
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	261.55	99.65	278.78	235.98
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	188.29	42.79	255.05	907.22
Nutrient balance	73.26	56.86	23.73	-671.24
Applied to removed ratio	1.39	2.33	1.09	0.26

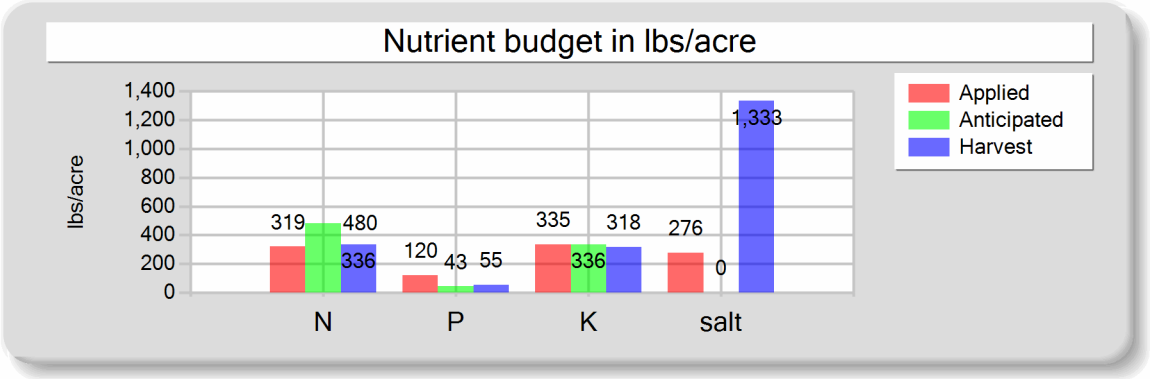
Fresh water applied	
37,904,617.00	*gallons*
1,395.90	*acre-inches*
22.16	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

12 - 11/06/2018: Alfalfa, hay

Field name: 12 Crop: Alfalfa, hay Plant date: 11/06/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	304.26	119.58	334.53	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.17	0.00	0.00	276.07
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	319.43	119.58	334.53	276.07
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	335.57	54.59	317.91	1,332.65
Nutrient balance	-16.14	64.99	16.62	-1,056.57
Applied to removed ratio	0.95	2.19	1.05	0.21

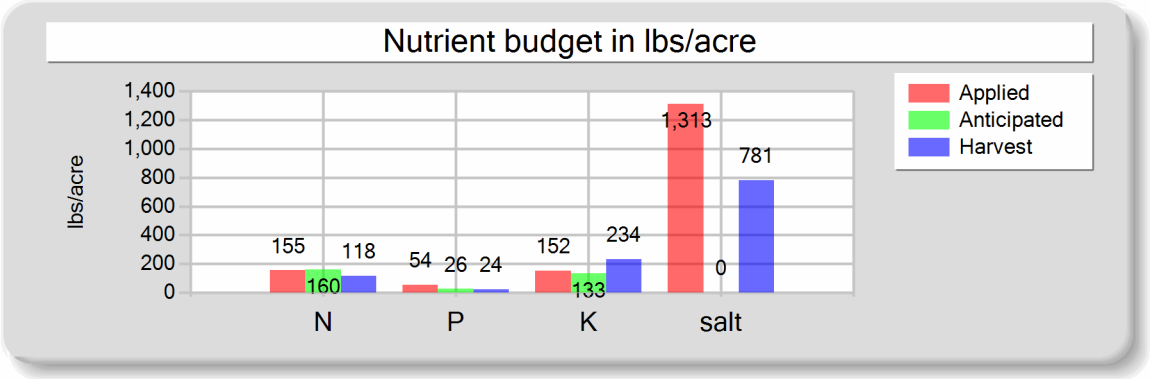
Fresh water applied	
24,635,965.00	*gallons*
907.26	*acre-inches*
25.92	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
7	*harvests*

13 - 11/23/2022: Oats, silage-soft dough

Field name: 13 Crop: Oats, silage-soft dough Plant date: 11/23/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	138.04	54.26	151.78	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.77	0.00	0.00	1,312.79
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	154.82	54.26	151.78	1,312.79
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	118.44	23.94	234.36	781.20
Nutrient balance	36.38	30.32	-82.58	531.59
Applied to removed ratio	1.31	2.27	0.65	1.68

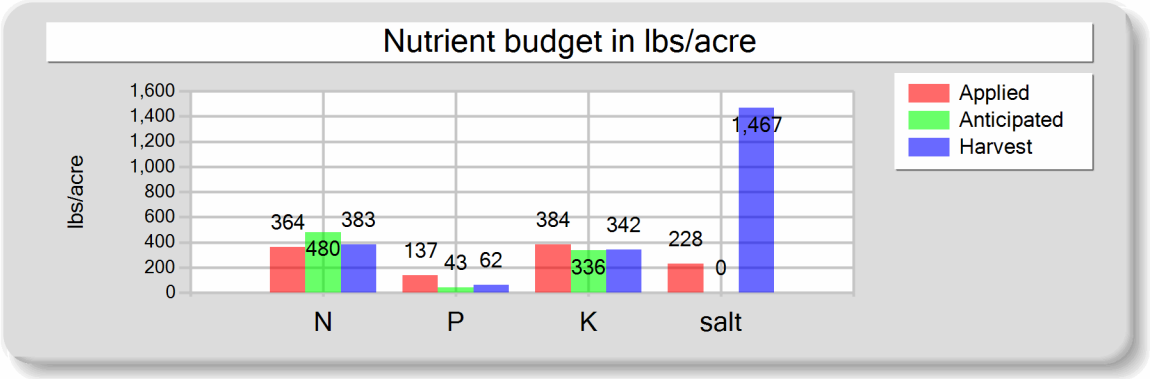
Fresh water applied	
47,093,562.00	*gallons*
1,734.30	*acre-inches*
96.35	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

14 - 11/11/2016: Alfalfa, hay

Field name: 14 Crop: Alfalfa, hay Plant date: 11/11/2016



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	349.05	137.19	383.78	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.97	0.00	0.00	228.43
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	364.02	137.19	383.78	228.43
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	382.80	62.40	342.33	1,467.13
Nutrient balance	-18.78	74.79	41.45	-1,238.70
Applied to removed ratio	0.95	2.20	1.12	0.16

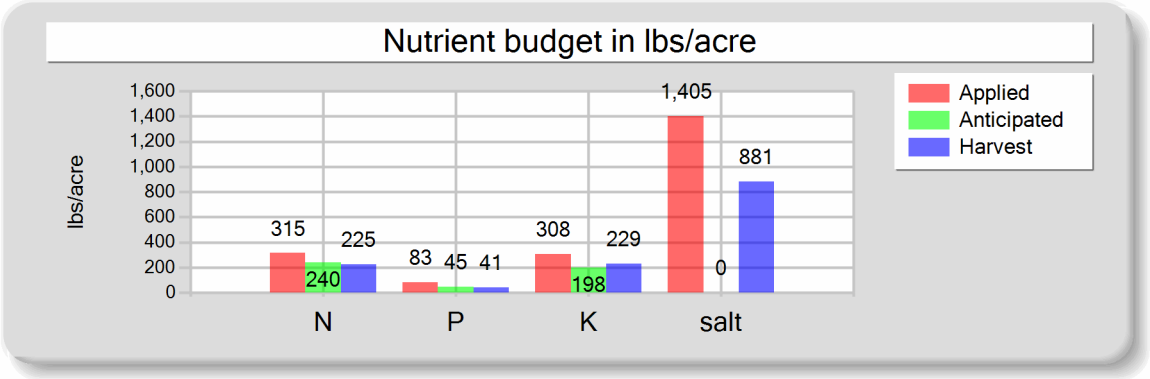
Fresh water applied
34,944,914.00 *gallons*
1,286.90 *acre-inches*
21.45 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
7 *harvests*

2 - 04/17/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 04/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	64.52	26.36	126.26	0.00
Process wastewater	235.19	56.49	181.32	1,069.35
Fresh water	1.12	0.00	0.00	335.56
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	314.82	82.85	307.59	1,404.91
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	225.43	40.68	228.82	881.39
Nutrient balance	89.39	42.17	78.77	523.52
Applied to removed ratio	1.40	2.04	1.34	1.59

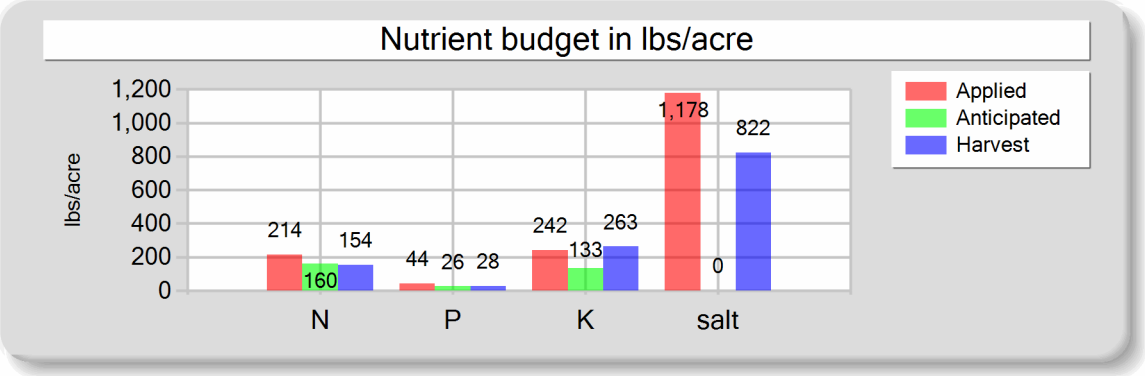
Fresh water applied	
32,528,243.00	*gallons*
1,197.90	*acre-inches*
28.52	*inches/acre*

Process wastewater applied	
1,170,000.00	*gallons*
43.09	*acre-inches*
1.03	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 11/21/2022: Oats, silage-soft dough

Field name: 7 Crop: Oats, silage-soft dough Plant date: 11/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	27.97	11.43	54.73	0.00
Process wastewater	179.15	32.47	187.55	1,082.08
Fresh water	0.25	0.00	0.00	96.08
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	214.36	43.90	242.28	1,178.16
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	153.70	28.31	262.90	822.41
Nutrient balance	60.67	15.59	-20.63	355.74
Applied to removed ratio	1.39	1.55	0.92	1.43

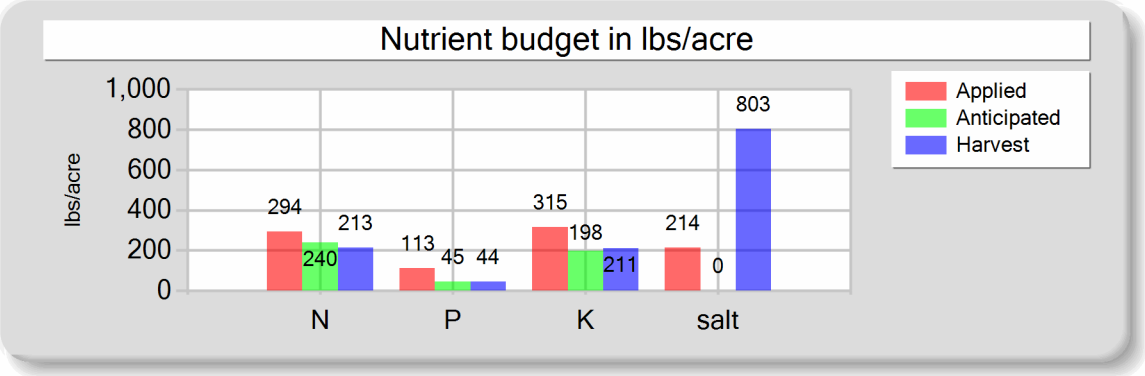
Fresh water applied	
11,563,822.00	*gallons*
425.86	*acre-inches*
7.47	*inches/acre*

Process wastewater applied	
1,755,000.00	*gallons*
64.63	*acre-inches*
1.13	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 05/19/2023: Corn, silage

Field name: 7 Crop: Corn, silage Plant date: 05/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	286.46	112.59	314.97	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.91	0.00	0.00	213.88
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	294.37	112.59	314.97	213.88
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	213.09	44.26	211.45	803.19
Nutrient balance	81.28	68.33	103.52	-589.31
Applied to removed ratio	1.38	2.54	1.49	0.27

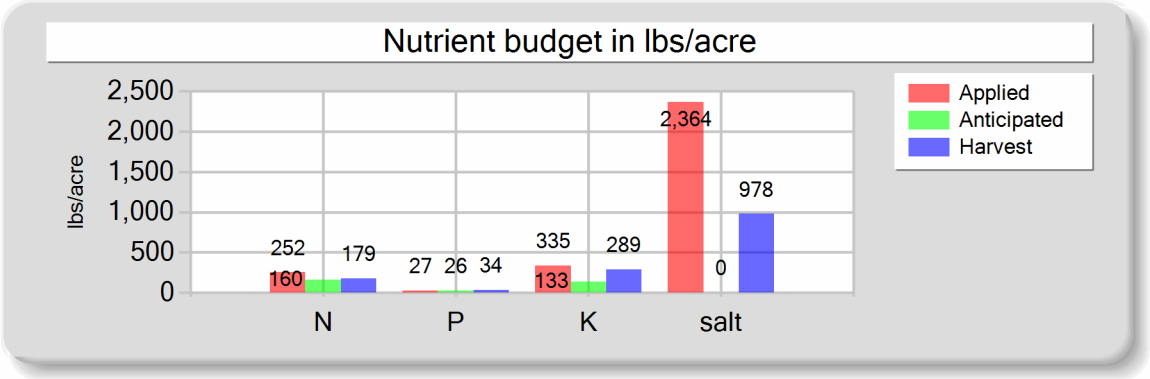
Fresh water applied	
31,082,682.00	*gallons*
1,144.67	*acre-inches*
20.08	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 11/21/2022: Oats, silage-soft dough

Field name: 8 Crop: Oats, silage-soft dough Plant date: 11/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	244.72	27.19	334.63	2,208.83
Fresh water	0.24	0.00	0.00	154.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	251.96	27.19	334.63	2,363.65
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	179.48	33.56	288.92	977.65
Nutrient balance	72.48	-6.38	45.71	1,386.00
Applied to removed ratio	1.40	0.81	1.16	2.42

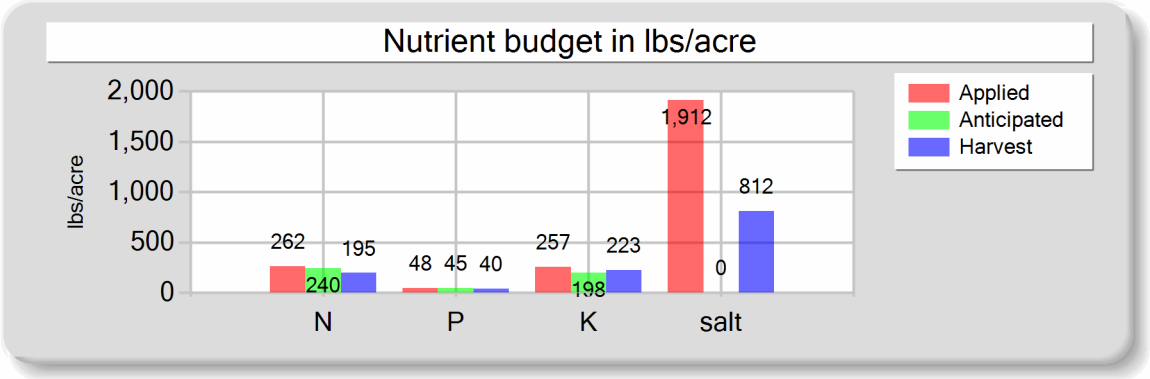
Fresh water applied	
17,424,867.00	*gallons*
641.70	*acre-inches*
10.52	*inches/acre*

Process wastewater applied	
3,450,000.00	*gallons*
127.05	*acre-inches*
2.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 05/19/2023: Corn, silage

Field name: 8 Crop: Corn, silage Plant date: 05/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	253.70	47.75	256.51	1,611.02
Fresh water	1.28	0.00	0.00	300.95
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	261.98	47.75	256.51	1,911.98
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	195.30	39.75	222.96	812.32
Nutrient balance	66.67	8.00	33.56	1,099.65
Applied to removed ratio	1.34	1.20	1.15	2.35

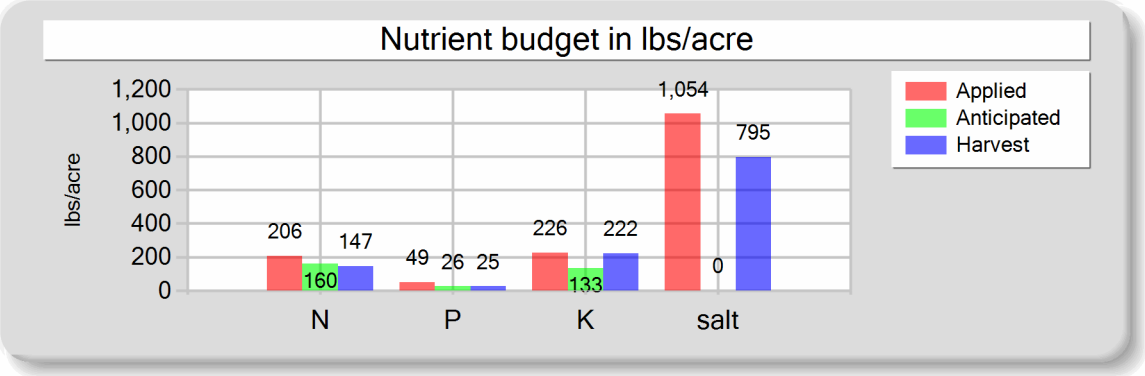
Fresh water applied	
46,806,183.00	*gallons*
1,723.71	*acre-inches*
28.26	*inches/acre*

Process wastewater applied	
2,535,000.00	*gallons*
93.36	*acre-inches*
1.53	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 11/22/2022: Oats, silage-soft dough

Field name: 9 Crop: Oats, silage-soft dough Plant date: 11/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	40.78	16.66	79.80	0.00
Process wastewater	158.56	32.82	146.55	903.47
Fresh water	0.09	0.00	0.00	150.50
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	206.42	49.48	226.35	1,053.98
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	147.43	25.34	222.29	794.73
Nutrient balance	59.00	24.15	4.06	259.25
Applied to removed ratio	1.40	1.95	1.02	1.33

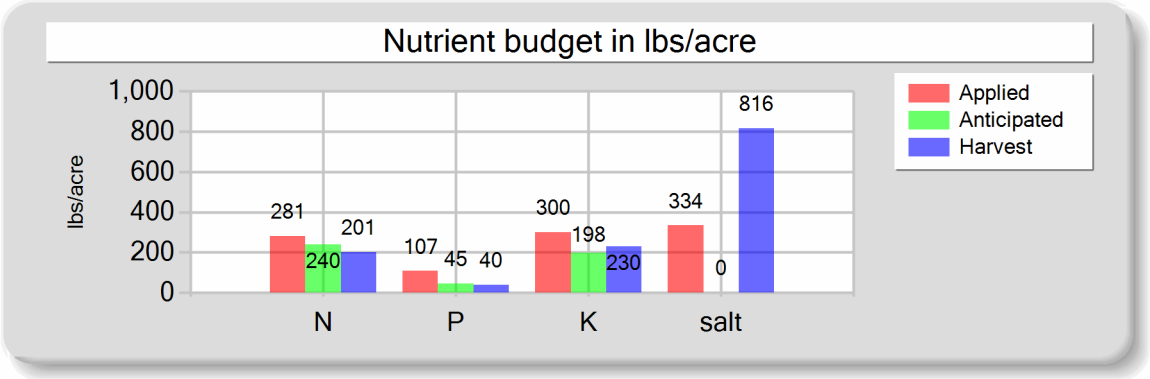
Fresh water applied	
10,343,824.00	*gallons*
380.93	*acre-inches*
8.86	*inches/acre*

Process wastewater applied	
1,005,000.00	*gallons*
37.01	*acre-inches*
0.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 05/29/2023: Corn, silage

Field name: 9 Crop: Corn, silage Plant date: 05/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	272.42	107.07	299.52	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.42	0.00	0.00	333.72
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	280.84	107.07	299.52	333.72
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	201.42	39.95	229.71	815.65
Nutrient balance	79.42	67.12	69.81	-481.93
Applied to removed ratio	1.39	2.68	1.30	0.41

Fresh water applied	
36,587,479.00	*gallons*
1,347.39	*acre-inches*
31.33	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure

Sample date: 05/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 11.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,100.00	7,900.00	22,100.00							
DL	100.00	100.00	100.00							

Solid Manure

Sample and source description: Solid Manure

Sample date: 11/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 14.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,300.00	3,800.00	18,200.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,010.00	403.00	0.00	1.70	243.00	780.00								9,740.00	4,600
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	577.00	390.00	0.00	1.10	83.50	707.00								8,970.00	3,700
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	518.00	364.00	0.00	0.50	57.60	709.00								8,300.00	4,680
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	582.00	340.00	0.00	1.10	60.70	742.00								9,010.00	5,380
DL	1.00	0.50	0.10	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - Dairy Well DW

Domestic Well - Dairy Well DW

Dairy Well DW

Sample description: Dairy Well DW

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.30	0.00	0.30								1,440.00	
DL	0.10	0.10	0.10								10.00	

Domestic Well - Dry Stock DW

Dry Stock DW

Sample description: Dry Stock DW

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.20	0.00	0.20								1,260.00	
DL	0.10	0.10	0.10								10.00	

Domestic Well - South Pump DW

South Pump DW

Sample description: South Pump DW

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	10.80	0.00	10.80								839.00	
DL	0.10	0.10	0.10								10.00	

Irrigation Water - Canal

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								33.30	83
DL	1.00	1.00	1.00								10.00	10

Irrigation Water - River Water

River Water

Sample description: River Water

Sample date: 08/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.20	0.00	0.20								37.80	47
DL	0.10	0.10	0.10								10.00	10

D. SOIL ANALYSES

1

1

Sample and source description: 1

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			38.10				
DL			1.00				

10

10

Sample and source description: 10

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			46.20				
DL			1.00				

11

11

Sample and source description: 11

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			67.20				
DL			1.00				

12

12

Sample and source description: 12

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			18.50				
DL			1.00				

14

14

14

Sample and source description: 14

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			18.50				
DL			1.00				

2

2

Sample and source description: 2

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			36.50				
DL			1.00				

3

3

Sample and source description: 3

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			29.90				
DL			1.00				

4

4

4

Sample and source description: 4

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			179.00				
DL			1.00				

5

5

Sample and source description: 5

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			25.70				
DL			1.00				

6

6

Sample and source description: 6

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			35.40				
DL			1.00				

7

7

7

Sample and source description: 7

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			92.30				
DL			1.00				

8

8

Sample and source description: 8

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			92.20				
DL			1.00				

9

9

Sample and source description: 9

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			42.80				
DL			1.00				

E. PLANT TISSUE ANALYSES

1 - 11/06/2018: Alfalfa, hay

1 - 11/06/2018: Alfalfa, hay

1

Sample and source description: 1

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 37.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	36,000.00	3,100.00	17,500.00		9.90
DL	10.00	10.00	10.00		0.01

1

Sample and source description: 1

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,100.00	3,400.00	18,500.00		7.60
DL	100.00	100.00	100.00		0.01

10 - 11/23/2022: Oats, silage-soft dough

10

Sample and source description: 10

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	2,300.00	19,200.00		7.70
DL	100.00	100.00	100.00		0.01

10 - 05/31/2023: Corn, silage

10 - 05/31/2023: Corn, silage

10

Sample and source description: 10

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	2,900.00	13,300.00		5.20
DL	100.00	100.00	100.00		0.01

11 - 11/23/2022: Oats, silage-soft dough

11A

Sample and source description: 11A

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	2,000.00	19,100.00		7.70
DL	100.00	100.00	100.00		0.01

11B

Sample and source description: 11B

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,600.00	1,900.00	18,600.00		7.90
DL	100.00	100.00	100.00		0.01

11 - 05/22/2023: Corn, silage

11 - 05/22/2023: Corn, silage

11A

Sample and source description: 11A

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,700.00	2,300.00	12,900.00		4.60
DL	100.00	100.00	100.00		0.01

11B

Sample and source description: 11B

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,000.00	2,500.00	14,900.00		5.30
DL	100.00	100.00	100.00		0.01

12 - 11/06/2018: Alfalfa, hay

12

Sample and source description: 12

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 39.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	39,500.00	3,200.00	19,100.00		9.30
DL	10.00	10.00	10.00		0.10

12 - 11/06/2018: Alfalfa, hay

12

Sample and source description: 12

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,900.00	3,400.00	19,800.00		8.30
DL	100.00	100.00	100.00		0.01

13 - 11/23/2022: Oats, silage-soft dough

13

Sample and source description: 13

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,400.00	1,900.00	18,600.00		6.20
DL	100.00	100.00	100.00		0.01

14 - 11/11/2016: Alfalfa, hay

14A

Sample and source description: 14A

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 37.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	37,200.00	3,100.00	17,900.00		9.80
DL	10.00	10.00	10.00		0.10

14 - 11/11/2016: Alfalfa, hay

14B

Sample and source description: 14B

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 38.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	38,100.00	3,100.00	17,500.00		9.80
DL	10.00	10.00	10.00		0.01

14A

Sample and source description: 14A

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,700.00	3,700.00	20,300.00		8.70
DL	100.00	100.00	100.00		0.01

14B

Sample and source description: 14B

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 9.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,900.00	3,600.00	20,200.00		8.80
DL	100.00	100.00	100.00		0.01

2 - 04/17/2023: Corn, silage

2 - 04/17/2023: Corn, silage

2

Sample and source description: 2

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,300.00	2,400.00	13,500.00		5.20
DL	100.00	100.00	100.00		0.01

7 - 11/21/2022: Oats, silage-soft dough

7A

Sample and source description: 7A

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	2,100.00	19,500.00		6.10
DL	100.00	100.00	100.00		0.01

7B

Sample and source description: 7B

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	2,200.00	20,400.00		6.80
DL	100.00	100.00	100.00		0.01

7 - 05/19/2023: Corn, silage

7 - 05/19/2023: Corn, silage

7A

Sample and source description: 7A

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,700.00	2,600.00	13,600.00		4.80
DL	100.00	100.00	100.00		0.01

7B

Sample and source description: 7B

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	2,700.00	12,900.00		4.90
DL	100.00	100.00	100.00		0.01

8 - 11/21/2022: Oats, silage-soft dough

8A

Sample and source description: 8A

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,300.00	2,300.00	19,800.00		6.70
DL	100.00	100.00	100.00		0.01

8 - 11/21/2022: Oats, silage-soft dough

8B

Sample and source description: 8B

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	2,300.00	20,400.00		6.80
DL	100.00	100.00	100.00		0.01

8 - 05/19/2023: Corn, silage

8A

Sample and source description: 8A

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,200.00	2,100.00	13,700.00		4.90
DL	100.00	100.00	100.00		0.01

8B

Sample and source description: 8B

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	2,300.00	12,900.00		4.70
DL	100.00	100.00	100.00		0.01

9 - 11/22/2022: Oats, silage-soft dough

9 - 11/22/2022: Oats, silage-soft dough

9

Sample and source description: 9

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,800.00	2,200.00	19,300.00		6.90
DL	100.00	100.00	100.00		0.01

9 - 05/29/2023: Corn, silage

9

Sample and source description: 9

Sample date: 09/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	2,400.00	13,800.00		4.90
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

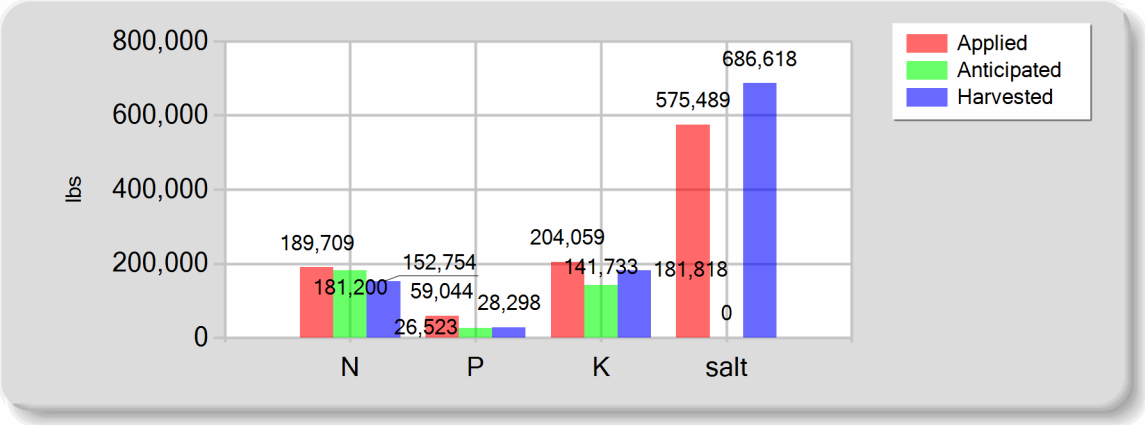
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

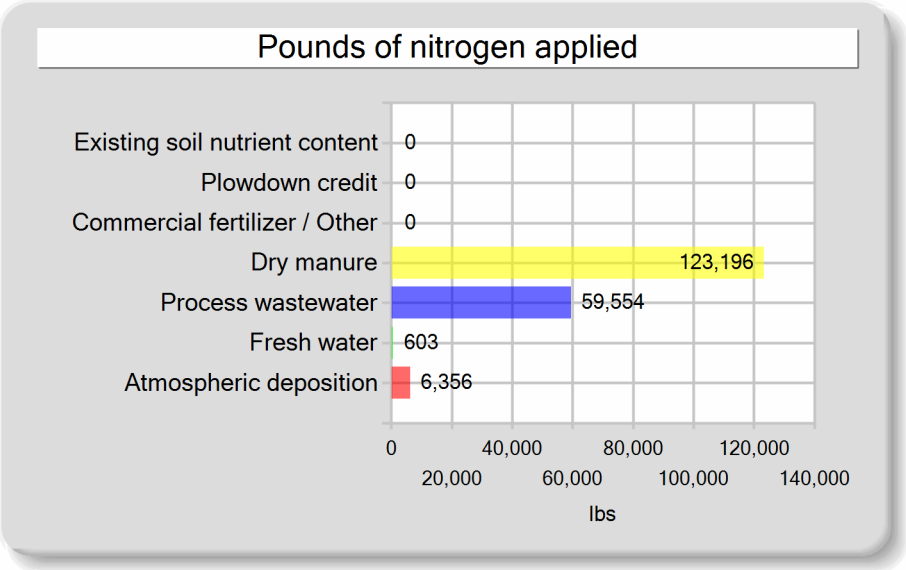
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

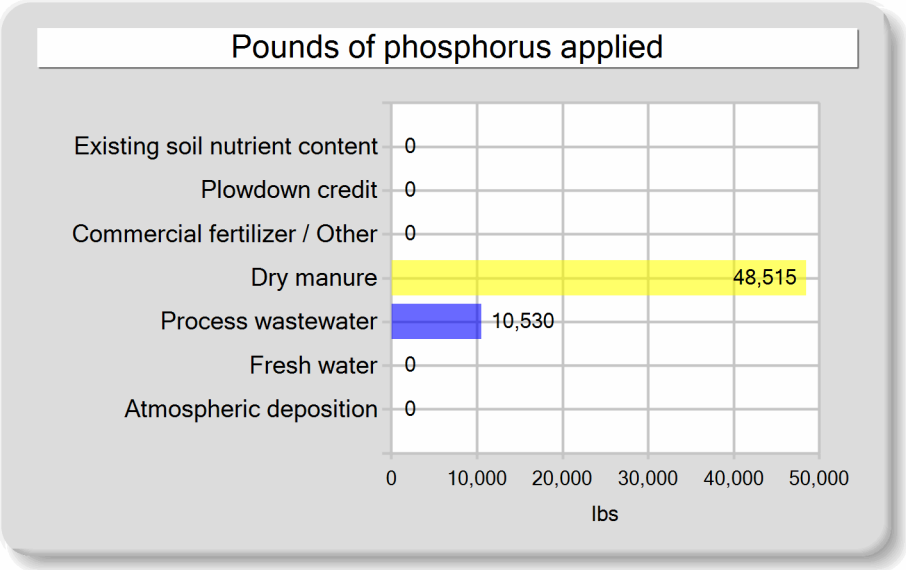
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	123,196.06	48,514.64	140,648.40	0.00
Process wastewater	59,553.86	10,529.80	63,410.78	392,809.16
Fresh water	603.21	0.00	0.00	182,680.04
Atmospheric deposition	6,356.00	0.00	0.00	0.00
Total nutrients applied	189,709.12	59,044.43	204,059.18	575,489.20
Anticipated crop nutrient removal	181,200.00	26,523.00	141,733.20	0.00
Actual crop nutrient removal	152,753.75	28,298.21	181,818.05	686,617.70
Nutrient balance	36,955.37	30,746.23	22,241.13	-111,128.50
Applied to removed ratio	1.24	2.09	1.12	0.84

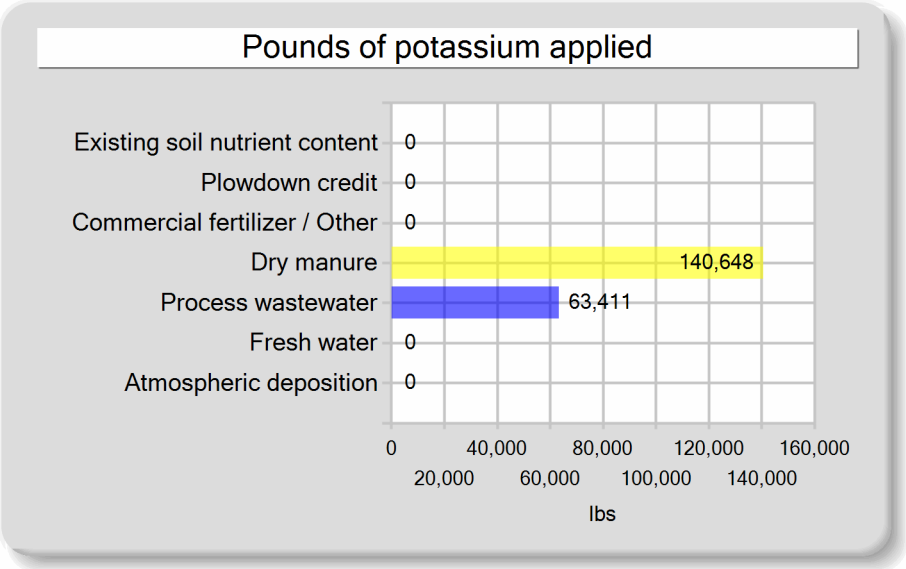
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

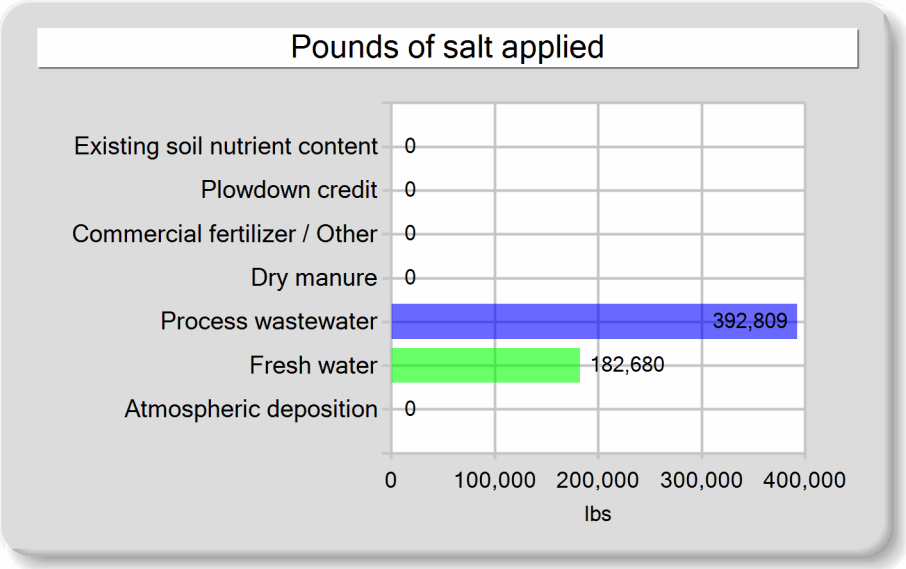


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Joe Souza	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-25-24	6-25-24
DATE	DATE

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: **Joe Souza**

Name of Dairy Facility: **Souza Brothers Dairy**

Facility Address: **3887 McCullagh Road, Stevinson, CA 95374**

Contact Person Name and Phone Number: **Joe or Dan Souza (209) 634-7522**

Manure Hauler Information:

Name of Hauling Company/Person: **Dores Ag Services, Inc.**

Address of Hauling Company /Person: **22846 Second Avenue, Stevinson, CA 95374**

Contact Person and Phone Number: **Phil Dores (209) 649-4503**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Farmer** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
Nelson Dairy	19481 Geer Road	Hilmar	95324	(209) 678-

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
19481 Geer Road	Hilmar	95324	Merced County

Dates Hauled: **5/17/2023**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: **1,100** Tons

Manure Solids Content (if amount reported in tons): **88.3%**

Method used to determine amount of manure: _____

Number of Truckloads x Average Truck Weight

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **N/A** Gallons

Method used to determine volume of process wastewater: **N/A**

Written Agreement:

****NOT NECESSARY FOR SOLID MANURE MANIFEST****

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ Date: 5-17-2023

Hauler's Signature: _____ Date: 5-17-2023

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Souza Brothers Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 08:59

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0892-01	Dairy Well DW	Ag Water	T. Madruga		11/15/2023 15:00
23K0892-02	Dry Stock DW	Ag Water	T. Madruga		11/15/2023 15:30
23K0892-03	South Pump DW	Ag Water	T. Madruga		11/15/2023 16:00

Default Cooler Temperature on Receipt °C: 0.5
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Souza Brothers Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 08:59

Sample Results

Sample: Dairy Well DW
23K0892-01 (Water)

Sampled: 11/15/2023 15:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.44**	mmhos/cm	0.01	1		11/20/23 13:12	SM 2510 B		BEK0615
Electrical Conductivity umhos	**1440**	umhos/cm	10.0	1		11/20/23 13:12	SM 2510 B		BEK0615
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 15:00	Field		BEK0552
Nitrate Nitrogen as NO3N	**0.3**	mg/L	0.1	1	10	11/16/23 21:15	EPA 300.0		BEK0603
Temperature	**25.0**	units	0.0	1		11/20/23 13:12	SM 4500-H+	H	BEK0615
pH	**8.2**	units	1.0	1		11/20/23 13:12	SM 4500-H+	H	BEK0615

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LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Souza Brothers Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 08:59

Sample Results
(Continued)

Sample: Dry Stock DW
23K0892-02 (Water)

Sampled: 11/15/2023 15:30
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.26**	mmhos/cm	0.01	1		11/20/23 13:14	SM 2510 B		BEK0615
Electrical Conductivity umhos	**1260**	umhos/cm	10.0	1		11/20/23 13:14	SM 2510 B		BEK0615
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 15:30	Field		BEK0552
Nitrate Nitrogen as NO3N	**0.2**	mg/L	0.1	1	10	11/16/23 21:37	EPA 300.0		BEK0603
Temperature	**25.0**	units	0.0	1		11/20/23 13:14	SM 4500-H+	H	BEK0615
pH	**8.2**	units	1.0	1		11/20/23 13:14	SM 4500-H+	H	BEK0615

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Souza Brothers Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 08:59

Sample Results
(Continued)

Sample: South Pump DW
23K0892-03 (Water)

Sampled: 11/15/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.84**	mmhos/cm	0.01	1		11/20/23 13:15	SM 2510 B		BEK0615
Electrical Conductivity umhos	**839**	umhos/cm	10.0	1		11/20/23 13:15	SM 2510 B		BEK0615
Ammonia (as N)	ND	mg/L	0.00	1		11/15/23 16:00	Field		BEK0552
Nitrate Nitrogen as NO3N	**10.8**	mg/L	0.1	1	10	11/16/23 21:59	EPA 300.0		BEK0603
Temperature	**25.0**	units	0.0	1		11/20/23 13:15	SM 4500-H+	H	BEK0615
pH	**7.6**	units	1.0	1		11/20/23 13:15	SM 4500-H+	H	BEK0615

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Received: 11/16/2023 15:35
Reported: 11/22/2023 08:59

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0603										
Blank (BEK0603-BLK1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0603-BLK2)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0603-BLK3)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0603-BS1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.2	90-110		
LCS (BEK0603-BS2)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	4.8		0.1	mg/L	5.000		96.8	90-110		
Duplicate (BEK0603-DUP1)		**Source: 23K0892-03**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	10.8		0.1	mg/L		10.8			0.157	10
Duplicate (BEK0603-DUP2)		**Source: 23K0898-05**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	25.4		0.1	mg/L		24.8			2.19	10
Matrix Spike (BEK0603-MS1)		**Source: 23K0892-03**			Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	15.9		0.1	mg/L	5.000	10.8	101	90-110		
Matrix Spike (BEK0603-MS2)		**Source: 23K0898-05**			Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	30.4		0.1	mg/L	5.000	24.8	112	90-110		
Reference (BEK0603-SRM1)					Prepared & Analyzed: 11/16/2023					
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		97.9	90-110		
Reference (BEK0603-SRM2)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	9.7			mg/L	10.00		97.1	90-110		
Reference (BEK0603-SRM3)					Prepared: 11/16/2023 Analyzed: 11/17/2023					
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		97.9	90-110		

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Ranch: Souza Brothers Dairy

Received: 11/16/2023 15:35
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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0615										
Blank (BEK0615-BLK1)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.7		1.0	units						
Blank (BEK0615-BLK2)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.4		1.0	units						
Blank (BEK0615-BLK3)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.0		1.0	units						
Duplicate (BEK0615-DUP1)		**Source: 23K0888-03**			Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1.78		0.01	mmhos/cm		1.78			0.477	10
pH	7.4		1.0	units		7.4			0.271	10
Electrical Conductivity umhos	1780		10.0	umhos/cm		1780			0.477	10
Duplicate (BEK0615-DUP2)		**Source: 23K0892-03**			Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	0.84		0.01	mmhos/cm		0.84			0.262	10
Electrical Conductivity umhos	841		10.0	umhos/cm		839			0.262	10
pH	7.6		1.0	units		7.6			0.00	10
Reference (BEK0615-SRM1)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	550			umhos/cm	538.0		102	90-110		
Reference (BEK0615-SRM2)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	7.6			units	7.520		101	67021-101.32		
Reference (BEK0615-SRM3)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEK0615-SRM4)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEK0615-SRM5)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		

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Ranch: Souza Brothers Dairy

Received: 11/16/2023 15:35
Reported: 11/22/2023 08:59

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0615 (Continued)										
Reference (BEK0615-SRM5)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEK0615-SRM6)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEK0615-SRM7)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEK0615-SRM8)					Prepared: 11/17/2023	Analyzed: 11/20/2023				
pH	4.0			units	4.000		101	97.5-102.5		

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11/16/23 15:35 23K0892

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 3 No. of Bottles:

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com

COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT Souza Brothers Dairy
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents

Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	Dairy Well DW	11/15/2023	1500	.5	ND
2	Dry Stock DW	11/15/2023	1530	.5	ND
3	South Pump DW	11/15/2023	1600	.4	ND
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/15/2023 @ 1600	11/16/2023 @9am
Second	JCF	Dellavalle Laboratory, Inc.	11/16/2023 @9am	11-16-23 3:25
Third		DLI	11.16.23 15:25	
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/16/23 15:35 23K0892

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☑ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☑ None ☐ Box ☐

Refrigerant: Wet Ice: ☑ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1	1							
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										